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- We introduce *Obhoy*, a permissioned claims-integrity ledger that *redefines* the insurable event as a single-use digital asset to *make* insurance verifiable, honest, and affordable.

[illegible]

Obhoy breaks the above loop at its root and turns it into a growth engine by zero-cost verification strips fraud margins from premiums, making policies affordable and restoring the trust needed to scale adoption

The left chart, titled 'Life (annual)', shows the percentage of claims settled for life insurance policies from 2020 to 2024. The percentages are 85% in 2020, 68% in 2021, 72% in 2023, and 66% in 2024. The right chart, titled 'General & industry (trly)', shows the percentage of claims settled for general and industry policies for the quarters Q3'25, Q4'25, and Q1'26. The percentages are 7.6% for Q3'25, 9.4% for Q4'25, and 27.1% for Q1'26.

Period	Claims settled (%)
2020	85%
2021	68%
2023	72%
2024	66%

Period	Claims settled (%)
Q3'25	7.6%
Q4'25	9.4%
Q1'26	27.1%

General & Industry

catastrophic systemic collapse, general insurers settling fewer than 1 in 10 filed claims by Q4 2025.

The diagram illustrates the IDRA system architecture, organized into layers and components:

- ACTORS:** Policy-holder, Provider(payee), Clinical verifier, Field agent, Insurer, IDRA, Academic auditor.
- END-USER L1:** Policyholder app + USSD/SMS, Provider assenter app, Verifier mobile (Idemix), Insurer dashboard, Regulator & auditor dashboards.
- APPLICATION L2:** API gateway (REST / gRPC), Chaincode invocation SDK, Settlement adapter (insurer core), Receipt & notification.
- MIDDLEWARE L3:** Idemix auth + role mapping, KYC / enrolment (NID-bound), Off-chain anomaly scoring, Domain profile key · attestors · settlement.
- BLOCKCHAIN Hyperledger Fabric L4:**
 - CHAINCODE:** Claim path IdentityRegistry · PolicyRegistry · EventRegistry BenefitSchedule · ClaimSettlement · ProviderRegistry; Oversight TransparencyLedger · GovernanceCouncil invariants: single-open-event, 2-of-3 quorum, single-use entitlement.
 - NETWORK:** Endorsing peers (one per class), Raft ordering O1–O5, Fabric MSP / CAs role-based access, Endorsement policy AND (OutOf (2, P, C, V), InsurerMSP).
 - Channels: main (policy, event, settlement state; private data collections per insurer) + audit (aggregates only).
- DATA L5:** On-chain state commitments, Private data collections, Encrypted off-chain store, Public explorer aggregates.
- TRUST BOUNDARY:** Separates the internal system from external rails.
- EXTERNAL rails:** National ID verification (A2), Hospital HMIS / DGHS registry, bKash / Nagad / Rocket, Insurer core / IDRA reporting, Public chain anchor Polygon / OTS.

Figure 1 is a line graph showing the cumulative share of weight (Y-axis) versus the cumulative share of stakeholder classes (X-axis). The Y-axis ranges from 0 to 1, and the X-axis ranges from 0 to 1. Three lines are plotted:

- Line of equality:** A dashed grey line representing perfect equality, where the cumulative share of weight equals the cumulative share of stakeholder classes.
- Obhoy ($n=5$, Gini 0.14):** A solid green line representing the Obhoy distribution. It is the uppermost curve, indicating the highest inequality.
- Reference ($n=3$, Gini 0.26):** A solid red line representing the reference distribution. It is the lowermost curve, indicating the lowest inequality.

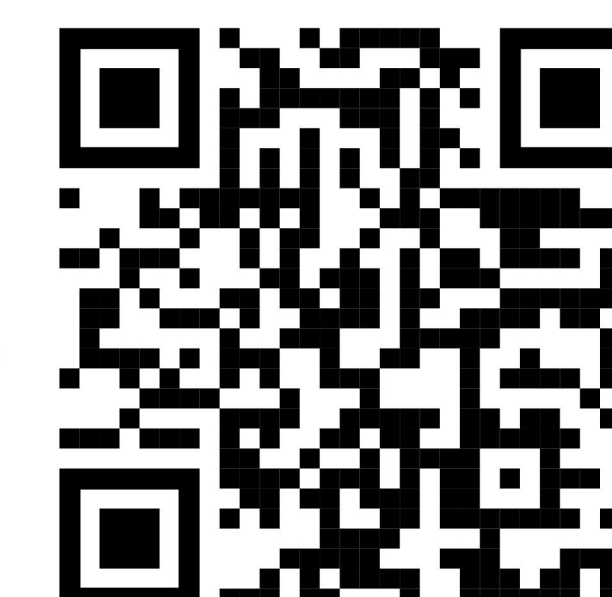
The Obhoy distribution is more unequal than the reference distribution, as its curve is further from the line of equality.

gap narrows year on year
as fixed cost is amortised

Year	Platform revenue (Tk crore)	Operating cost (Tk crore)
Year 1	0.2	1.2
Year 2	0.4	1.6
Year 3	0.9	2.0
Year 4	1.8	2.4

Implementation

Our code is publicly available!



```
graph TD; A[More providers accredited] --> B[Fraud & duplication fall]; B --> C[Predictable loss ratio]; C --> D[Lower premium, same solvency]; D --> E[Higher uptake, larger pool]; E --> A;
```

entry point: verification

– Conclusion

- Where buyers cannot trust an insurer's promise and insurers cannot trust a claimed loss, **Obhoy's** shared, multi-party is met with a reliable safety net, not a financial freefall.
- Beyond the ledgers and loss ratios, **Obhoy** exists for one deeply human reason: to ensure that a family's hardest moment ledger remains the only route to rebuilding that trust and protecting vulnerable households from financial ruin.